

Kyle Stein, Ph.D.

352-540-1331 | kylestein18@gmail.com | www.linkedin.com/in/stein-kyle

SUMMARY

Assistant Research Scientist at the University of West Florida's Center for Cybersecurity and Artificial Intelligence, specializing in data-efficient, robust AI for cybersecurity. My research focuses on developing label-efficient, generalizable cyber-AI defenses that remain accurate under distribution shift and are rigorously evaluated. Specific work includes transformer-based packet modeling with self-supervised pretraining and few-shot adaptation for enterprise and IoT malware detection; open-world backdoor defenses using vision-language models with compositional zero-shot reasoning; and continual/few-shot incremental learning with parameter-efficient Vision Transformer fine-tuning that preserves prior knowledge. Accomplishments include 10+ first-author publications, including ICCV, ICIP, and MILCOM, as well as teaching and applied research experience through Johns Hopkins University and APL, and recognition as an NSF CyberCorps SFS Scholar.

EDUCATION

University of West Florida

Ph.D. in Intelligent Systems and Robotics

Pensacola, FL

Aug. 2022 – May 2026

Advised By: Dr. Andrew A. Mahyari & Dr. Guillermo Francia III

Dissertation Title: Integrating Transformers for Cyber Defense Under Unseen and Evolving Threats with Deep Packet Inspection, Zero-Shot Backdoor Detection, and Continual Learning

University of West Florida

M.S. in Mathematical Sciences

Pensacola, FL

Aug. 2019 – May 2021

University of South Florida

B.S. in Interdisciplinary Natural Sciences

Tampa, FL

Aug. 2015 – Dec. 2018

TEACHING EXPERIENCE

Co-Instructor

Johns Hopkins University, Whiting School of Engineering

Baltimore, MD

May 2025 – Aug. 2025

- AI for Cybersecurity (EN.695.631.81), Summer 2025 (Co-Instructor)

Graduate Teaching Assistant & Lead Instructor

University of West Florida, Hal Marcus College of Science and Engineering

Pensacola, FL

Aug. 2019 – May 2021

- Biostatistics (STA4173), Spring 2021 (TA)
- College Algebra (MAC1105), Fall 2020 (Instructor)
- Applied Statistics (STA3162), Summer 2020 (TA)
- Elements of Statistics (STA2023), Summer 2020 (TA)
- Precalculus with Trigonometry (MAC1147), Fall 2019 (TA)

PROFESSIONAL EXPERIENCE

The University of West Florida

Pensacola, FL

Center for Cybersecurity and Artificial Intelligence, Assistant Research Scientist

July 2026 – Current

- Conduct research in trustworthy artificial intelligence and cybersecurity with an emphasis on AI-driven cyber defense.
- Develop and lead research initiatives focused on securing critical infrastructure and emerging cyber-physical systems.
- Prepare competitive research proposals and contribute to externally funded projects supported by federal agencies.
- Collaborate with academic, government, and industry partners to advance applied cybersecurity and AI research.

Johns Hopkins University Applied Physics Laboratory

Laurel, MD

Critical Infrastructure Protection Group, Internship

May 2024 – Aug. 2025

- Developed an LLM-based reinforcement-learning predictive assistant agent for active cyber-threat detection.
- Investigated machine-learning techniques for detecting cyberattacks on a hydroelectric-dam testbed.
- Contributed to the design and build of the physical hydroelectric-dam testbed hardware.
- Enhanced the MOSAICS framework for network-wide intrusion detection and alerting.
- Designed unit testing for data ingestion and validation pipelines to support a team of 25+ engineers.

U.S. Department of Homeland Security

Remote (Washington, DC)

Science & Technology Directorate, Research Internship

May 2023 – Aug. 2023

- Conducted predictive analyses on Data Analytics, Modeling & Simulation, Cybersecurity, AI/ML, and Quantum Information Systems over a five-year horizon.
- Proposed strategies for integrating explainable AI (XAI) into DHS operational components.
- Performed research on technical challenges present among DHS components and proposed component specific solutions.

PUBLICATIONS

Journal Publications

1. Stein K., Mahyari A., Francia G., & El-Sheikh E. **Packet Inspection Transformer: A Self-Supervised Journey to Unseen Malware Detection with Few Samples.** IEEE Access, 2025.
2. Stein K., Mahyari A., Francia G., & El-Sheikh E. **Proactive Disentangled Modeling of Trigger–Object Pairings for Backdoor Defense.** Computers, Materials & Continua, 2025.

Refereed Conference Publications

1. Stein K., Mahyari A., Francia G., & El-Sheikh E. **LSA-PAST: Layer-Specific Adaptation and Prototype Transport for Compositional Incremental Learning.** (Submitted).
2. Stein K., Mahyari A., Francia G., & El-Sheikh E. **Adapter-Based Few-Shot Continual Learning for Malicious Packet Recognition.** In Proceedings of IEEE Military Communications Conference (MILCOM), Washington, DC, USA, Oct. 12 – Oct. 16, 2026.
3. Stein K., Mahyari A., Francia G., & El-Sheikh E. **Visual Adaptive Prompting for Compositional Zero-Shot Learning.** Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV), Honolulu, HI, USA, Oct. 19–23, 2025.
4. Stein K., Mahyari A., Francia G., & El-Sheikh E. **Transductive One-Shot Learning Meets Subspace Decomposition.** Proceedings of the IEEE International Conference on Image Processing (ICIP), Anchorage, AK, USA, Sep. 14–17, 2025.
5. Stein K., Mahyari A., Francia G., & El-Sheikh E. **Adaptive Additive Parameter Updates of Vision Transformers for Few-Shot Continual Learning.** 59th IEEE Asilomar Conference on Signals, Systems, and Computers, Pacific Grove, CA, USA, Oct. 26–29, 2025.
6. Stein K., Mahyari A., Francia G., & El-Sheikh E. **Prompt Conditioned Vision-Language Models for Detecting Novel Backdoor Images.** 59th IEEE Asilomar Conference on Signals, Systems, and Computers, Pacific Grove, CA, USA, Oct. 26–29, 2025.
7. Stein K., Price A., Donato J., Cui M., Garcia H., & Watkins L. **From Packets to Pixels: A Vision Transformer & Few-Shot Learning Approach to Malicious Network Threat Detection.** In Proceedings of IEEE Military Communications Conference (MILCOM), Los Angeles, CA, USA, Oct. 6 – Oct. 10, 2025.
8. Stein K., Mahyari A., Francia G., & El-Sheikh E. **Proactive Adversarial Defense: Harnessing Prompt Tuning in Vision-Language Models to Detect Unseen Backdoored Images.** In Proceedings of IEEE Military Communications Conference (MILCOM), Los Angeles, CA, USA, Oct. 6 – Oct. 10, 2025.
9. Stein K., Mahyari A., Francia G., & El-Sheikh E. **Towards Novel Malicious Packet Recognition: A Few-Shot Learning Approach.** In Proceedings of IEEE Military Communications Conference (MILCOM), Washington, DC, USA, Oct. 28 – Nov. 1, 2024.
10. Stein K., Mahyari A., Francia G., & El-Sheikh E. **A Transformer-Based Framework for Payload Malware Detection and Classification.** 2024 IEEE World AIoT Congress (AIoT), Seattle, WA, USA, May 29–31, 2024.
11. Stein K., Mahyari A., & El-Sheikh E. **Vehicle Controller Area Network Inspection Using Recurrent Neural Networks.** International Conference on Advances in Computing Research (ICR), Orlando, FL, USA, May 8–10, 2023.
12. Stein K., Harvey A., Lopez A., Taj U., Watkins S., & Watkins L. **Eliciting and Measuring Toxic Bias in Human-to-Machine Interactions in Large Language Models.** 2024 IEEE 15th Annual Ubiquitous Computing, Electronics & Mobile Communication Conference (UEMCON), New York, NY, USA, Oct. 17–19, 2024.

CONFERENCE PRESENTATIONS

1. Stein K. *Adaptive Additive Parameter Updates of Vision Transformers for Few-Shot Continual Learning.* Presented at the **IEEE 59th Asilomar Conference on Signals, Systems, and Computers**, Pacific Grove, CA, USA, Oct. 26–29, 2025.
2. Stein K. *Prompt Conditioned Vision-Language Models for Detecting Novel Backdoor Images.* Presented at the **IEEE 59th Asilomar Conference on Signals, Systems, and Computers**, Pacific Grove, CA, USA, Oct. 26–29, 2025.
3. Stein K. *Visual Adaptive Prompting for Compositional Zero-Shot Learning.* Presented at the **IEEE/CVF International Conference on Computer Vision (ICCV)**, Honolulu, HI, USA, Oct. 19–23, 2025.
4. Stein K. *Transductive One-Shot Learning Meets Subspace Decomposition.* Presented at the **IEEE International Conference on Image Processing (ICIP)**, Anchorage, AK, USA, Sep. 14–17, 2025.
5. Stein K. *Transformer-Based Framework for Payload Malware Detection and Classification.* Presented at the **IEEE World AIoT Congress (AIoT)**, Seattle, WA, USA, May 29–31, 2024.
6. Stein K. *Towards Novel Malicious Packet Recognition: A Few-Shot Learning Approach.* Presented at **IEEE Military Communications Conference (MILCOM)**, Washington, DC, USA, Oct. 28 – Nov. 1, 2024.
7. Stein K. *Eliciting and Measuring Toxic Bias in Human-to-Machine Interactions in Large Language Models.* Presented at the **2024 IEEE 15th Annual Ubiquitous Computing, Electronics & Mobile Communication Conference (UEMCON)**, IBM T.J. Watson Research Center, New York, NY, USA, Oct. 17–19, 2024.
8. Stein K. *Transformer-Based Framework for Payload Malware Detection and Classification.* Presented at the **IEEE World**

AIoT Congress (AIIoT), Seattle, WA, USA, May 29–31, 2024.

9. Stein K. *Vehicle Controller Area Network Inspection Using Recurrent Neural Networks*. Presented at the **International Conference on Advances in Computing Research (ICR)**, Orlando, FL, USA, May 8-10, 2023.

AFFILIATIONS

National Science Foundation, CyberCorps Scholarship for Service (SFS), Grant Awardee
IEEE Member

CORE SKILLS

Machine & Deep Learning Expertise: Transformers, RNN, CNN, DNN, Few-Shot Learning, Computer Vision
Programming & Tools: Python, R, MATLAB, GitLab, Tableau, PyTorch, TensorFlow